

BUSINESS REQUIREMENTS SPECIFICATION (BRS)

Project Name: Hallo Teller – Hologram Bank Teller System

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2. Executive Summary

2.1 Problem Statement

Banks continue to rely heavily on physical branch tellers to execute routine banking services such as cash deposits, withdrawals, account enquiries, card support, statement printing, and customer guidance. While these services are critical for customers, they often result in long queues, inconsistent service quality, high staffing requirements, and limited availability outside standard working hours. In addition, manual teller operations increase the risk of human error, operational fraud exposure, and compliance gaps. Customers increasingly demand a faster and more digital experience while still expecting human-like interaction and reassurance in a branch environment.

2.2 Proposed Solution

Hallo Teller is a hologram-based bank teller designed to replicate and enhance the full capabilities of a human teller. The hologram will interact with customers through natural voice communication, guiding them through transactions and banking requests. Hallo Teller will securely integrate with the Core Banking System (CBS), authentication services, and branch devices such as printers, scanners, cash deposit machines, and queue management systems. The solution will also include escalation mechanisms to allow customers to connect with a human teller or supervisor whenever required by policy, customer preference, or transaction complexity.

2.3 Project Vision

The vision is to modernize the banking branch experience by deploying Hallo Teller as the first point of contact for walk-in customers. This will enable consistent service delivery, reduce branch congestion, increase operational efficiency, and provide a premium customer experience. Hallo Teller will act as a scalable digital workforce layer for branches, operating with strong governance controls, compliance enforcement, and full audit traceability.

2.4 Expected Business Value

- Reduction in teller workload by approximately 60% through automation of routine tasks.
- Faster customer service turnaround and reduced queue waiting time by at least 50%.
- Improved operational compliance through automated enforcement of banking rules and audit logs.
- Increased customer satisfaction and enhanced brand image through innovative service delivery.

- Reduced operational costs related to staffing, overtime, and repetitive manual activities.

3. Scope Definition

3.1 In Scope

- Voice-based customer interaction and conversational support.
- Customer authentication through OTP, biometric verification, or ID scanning.
- Integration with Core Banking System for transaction processing and customer data retrieval.
- Support for key teller transactions including balance inquiry, transfers, and deposits.
- Receipt printing, transaction confirmation, and customer notification support.
- Escalation workflow to human teller or supervisor when required.

3.2 Out of Scope

- Fully autonomous lending and credit approval decisions without human authorization.
- Direct physical cash dispensing from the hologram unit (requires ATM/CDM systems).
- Unrestricted execution of high-risk corporate banking transactions without compliance checks.

4. Functional Requirements

Req ID	Requirement	Priority
FR-01	Greet and guide customers through voice-based interaction.	High
FR-02	Support multilingual interaction (minimum 3 languages).	High
FR-03	Authenticate customers via OTP and/or biometric verification.	High
FR-04	Provide balance inquiry and account information retrieval.	High
FR-05	Provide mini-statement display and statement printing.	High
FR-06	Support cash deposit workflow through CDM integration.	High
FR-07	Support fund transfers between customer accounts.	High
FR-08	Support bill payments and scheduled payment confirmation.	Medium
FR-09	Support card replacement and card request initiation.	Medium
FR-10	Escalate customers to a human teller or supervisor via branch alert/video call.	High
FR-11	Detect fraud triggers and block suspicious transactions automatically.	High
FR-12	Log all customer actions, authentication attempts, and transaction events.	High

5. Non-Functional Requirements

Performance Requirements

- Voice response time must be less than 3 seconds under normal branch load.
- Transaction processing time must be less than 10 seconds excluding approval workflows.

Security Requirements

- Mandatory authentication required before any transaction execution.
- TLS encryption must be enforced for all communication with backend systems.
- Sensitive transactions must require multi-factor verification.

Availability Requirements

- Hallo Teller system must provide 99.5% uptime per month.
- The system must fail safely and escalate to human staff during downtime.

Disaster Recovery Requirements

- Recovery Time Objective (RTO): 4 hours.
- Recovery Point Objective (RPO): 15 minutes.